

Log Scale Surprise Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	log-scale-surprise

Big idea: Mathematical representations can make the same relationship look explosive, steady, or manageable.

One-Minute Teacher Brief

Students compare linear and logarithmic graphs and see how scale choice changes visual interpretation.

Student Prompt

Inspect Log Scale Surprise Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Log Scale Surprise Stimulus A	Student-facing stimulus 1: Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Log Scale Surprise Stimulus B	Student-facing stimulus 2: Question: does the graph show runaway growth, steady multiplicative growth, or both?. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Log Scale Surprise Reveal / Twist	Student-facing stimulus 3: Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical representations can make the same relationship look explosive, steady, or manageable.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Log Scale Surprise Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Log Scale Surprise Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does mathematical representation shape understanding?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical representations can make the same relationship look explosive, steady, or manageable.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: How does mathematical representation shape understanding?

Setup Checklist

- Use simple doubling data so students can see both the numerical and visual relationship.
- Let students describe each graph before teaching the term logarithmic scale.

Example Stimuli or Materials

- Dataset: 1, 2, 4, 8, 16, 32, 64 over seven days.
- Question: does the graph show runaway growth, steady multiplicative growth, or both?
- Notation card: $\log_{10}(1000) = 3$ shows how a multiplicative change becomes an equal step on a log scale.

Resources To Use

- Resource pack: <resources/log-scale-surprise/index.html>
- Card deck CSV: <resources/log-scale-surprise/cards.csv>
- Visual prompt: <resources/log-scale-surprise/visual-prompt.svg>
- Linear and logarithmic graph templates for the same dataset.
- Graph purpose cards: communicate urgency, compare growth rates, inspect proportional change.

Run Sequence

1. Give students a simple growing dataset.
2. Groups plot or inspect the same data on a linear graph and a logarithmic graph.
3. Students describe the story each graph seems to tell.
4. Groups identify which graph is better for different purposes.

5. Debrief how mathematical representation affects knowledge claims.

Debrief Moves

- Knowledge question: How does mathematical representation shape understanding?
- Can one graph be more truthful than another if both are accurate?
- What responsibilities come with choosing a scale?

Teacher Quick Script

- Both graphs can be accurate, but they serve different questions.
- What does this scale make visually obvious, and what does it hide?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...